In [1]:
```
import matplotlib
%matplotlib inline

matplotlib.rc('figure', figsize=(12, 5))
font = {'family' : 'normal',
        'weight' : 'bold',
        'size'   : 16}

matplotlib.rc('font', **font)
```

In [2]:
```
# September 2018 - Luke Harold Miles - Public Domain
# Simulated annealing solution to traveling salesman problem
# adapted from https://gist.github.com/qpwo/a46274751cc5db2ab1d936980072a134
import numpy as np


def find_tour(cities, dist_fun, iterations=10**5):
    def calc_distance():
        """sum of distance to and from i and j in tour
        """
        return sum(
            dist_fun(
                cities[tour[k % n]],
                cities[tour[(k + 1) % n]]
            )
            for k in [j - 1, j, i - 1, i]
        )

    n = len(cities)
    # initial tour randomly initialized
    tour = np.random.permutation(n)
    lengths = []
    for temperature in np.logspace(4, 0, num=iterations):
        i = np.random.randint(n - 1) # city 1
        j = np.random.randint(i + 1, n) # city 2
        old_length = calc_distance()
        tour[[i, j]] = tour[[j, i]] # swap i and j in tour
        new_length = calc_distance()
        if np.exp((old_length - new_length) / temperature) < np.random.random():  # swap was mist
            tour[[i, j]] = tour[[j, i]]  # swap them back
            lengths.append(old_length)
        else:
            lengths.append(new_length)
    return tour, lengths
```

In [3]:
```
from matplotlib import pyplot as plt
%matplotlib inline

def plot_solution(cities, tour):
    xcoords, ycoords = list(zip(*cities[tour].tolist(), cities[tour[0]].tolist()))
    plt.plot(xcoords, ycoords, 'xb-')
    plt.show() # graph the tour
```

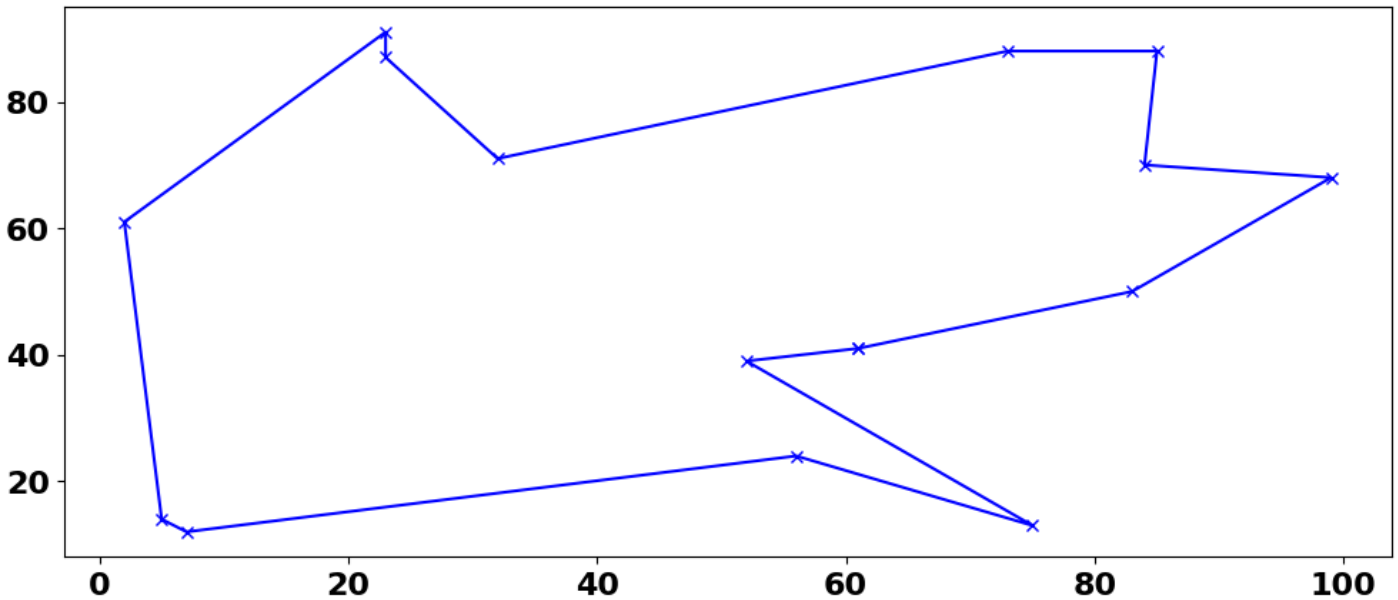
In [4]:
```
cities = np.random.randint(0, 100, (15, 2))
# scipy.spatial.distance.euclidean
def euclidean_distance(c1, c2):
    return np.linalg.norm(c1 - c2)
```

In [5]:
```
from scipy.spatial.distance import euclidean
```

In [6]:
```
tour, lengths = find_tour(cities, euclidean, iterations=1000000)
print(f'length: {lengths[-1]}')
```

```
length: 116.33514579533036
```

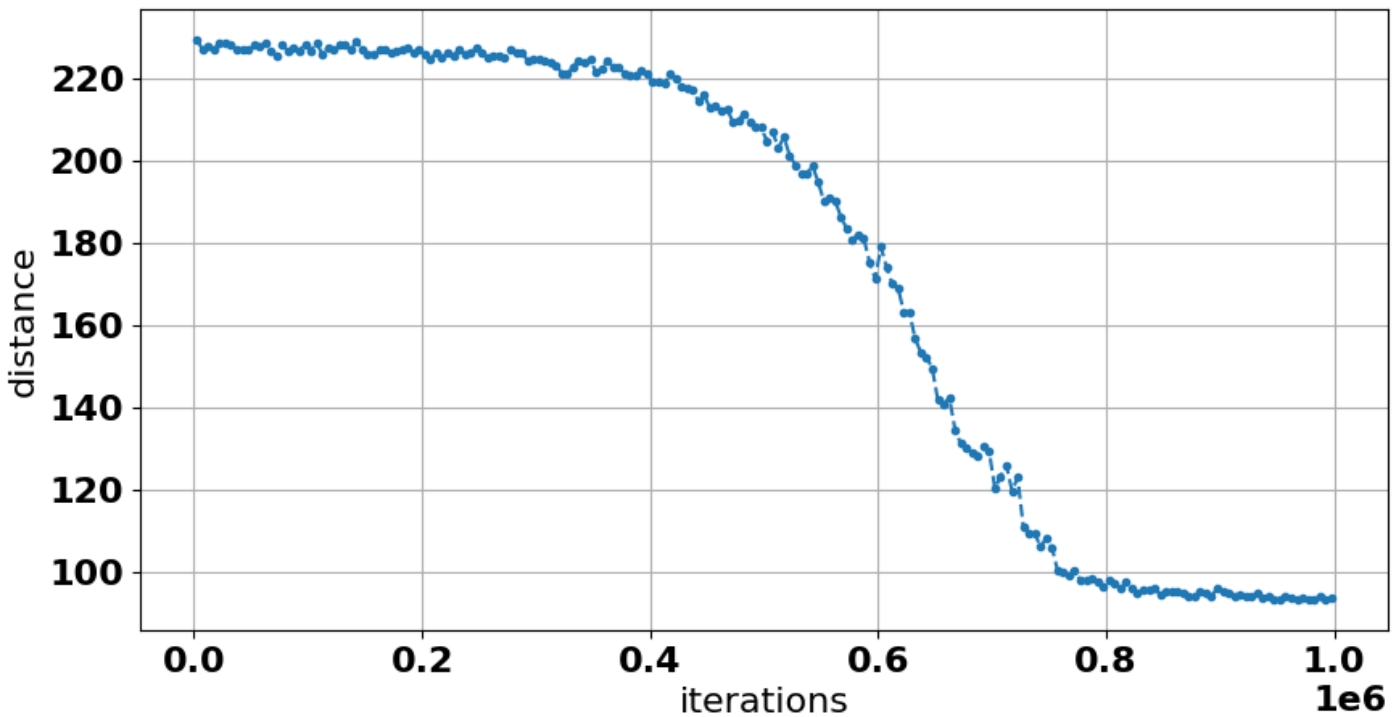
In [16]:
```
import matplotlib.pyplot as plt
plt.rcParams['font.family'] = 'DejaVu Sans'

plot_solution(cities, tour)
```



In [13]:
```
# plt.plot(t, t, 'r--', t, t**2, 'bs', t, t**3, 'g^')
from scipy.stats import sem
import matplotlib.pyplot as plt
plt.rcParams['font.family'] = 'DejaVu Sans'


# def plot_learning(iterations: int, lengths, disp_iter=1000):
#     bin_size = int(iterations // float(min([iterations, disp_iter])))
#     # errors = np.reshape(lengths, newshape=(-1, bin_size))
#     # iters = np.reshape(list(range(iterations)), newshape=(-1, bin_size))
#     errors = np.reshape(lengths, (-1, bin_size))
#     iters  = np.reshape(np.arange(iterations), (-1, bin_size))

#     plt.errorbar(
#         iters.mean(axis=1),
#         errors.mean(axis=1),
#         yerr=sem(errors, axis=1),
#         fmt='--o',
#     )
#     plt.ylabel('distance')
#     plt.xlabel('iterations')


def plot_learning(iterations: int, lengths, disp_iter=200):
    bin_size = iterations // disp_iter

    errors = np.reshape(lengths, (-1, bin_size))
    iters  = np.reshape(np.arange(iterations), (-1, bin_size))

    plt.figure(figsize=(10, 5))
```

```
    plt.errorbar(
        iters.mean(axis=1),
        errors.mean(axis=1),
        yerr=sem(errors, axis=1),
        fmt='--o',
        markersize=3
    )
    plt.ylabel('distance')
    plt.xlabel('iterations')
    plt.grid(True)
    plt.show()
```

In [15]:
```
import matplotlib.pyplot as plt
plt.rcParams['font.family'] = 'DejaVu Sans'

plot_learning(1000000, lengths)
```



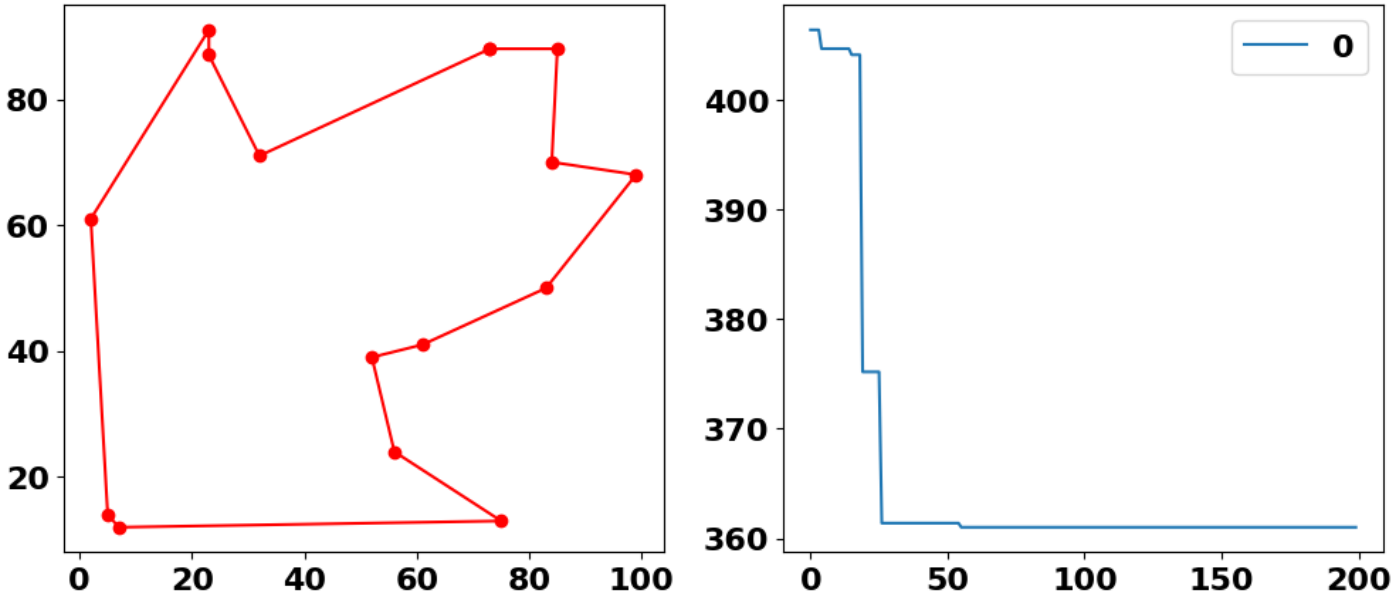
In [17]:
```
from scipy import spatial

distance_matrix = spatial.distance.cdist(cities, cities, metric='euclidean')
num_points = cities.shape[0]
```

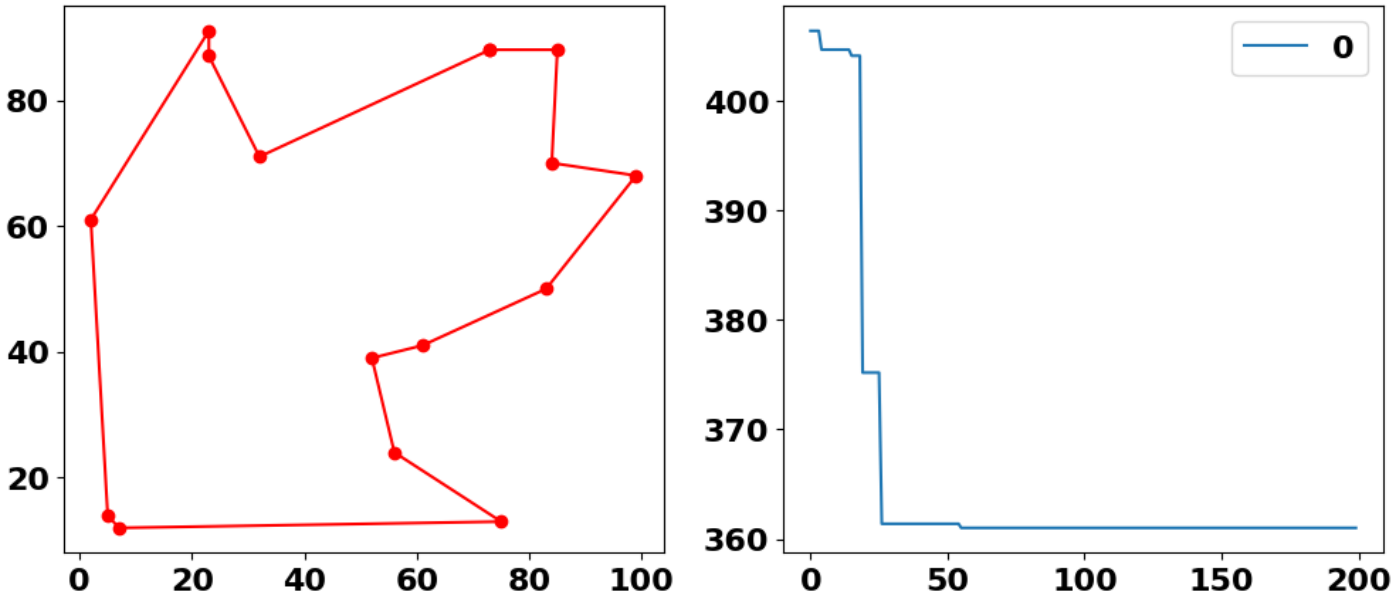
In [19]:
```
from sko.ACA import ACA_TSP
import numpy as np
np.int = int

N = len(distance_matrix)
def cal_total_distance(tour):
    '''The objective function. input tour, return total distance.
    cal_total_distance(np.arange(num_points))
    '''
    return sum([
        distance_matrix[tour[i % N], tour[(i + 1) % N]]
        for i in range(N)
    ])

aca = ACA_TSP(func=cal_total_distance, n_dim=N,
              size_pop=num_points, max_iter=200,
              distance_matrix=distance_matrix)
```

```
best_x, dist = aca.run()
```

In [20]:
```
import pandas as pd
```

In [24]:
```
fig, ax = plt.subplots(1, 2)
best_points_ = np.concatenate([best_x, [best_x[0]]])
best_points_coordinate = cities[best_points_, :]
ax[0].plot(best_points_coordinate[:, 0], best_points_coordinate[:, 1], 'o-r')
pd.DataFrame(aca.y_best_history).cummin().plot(ax=ax[1])
plt.show()
```





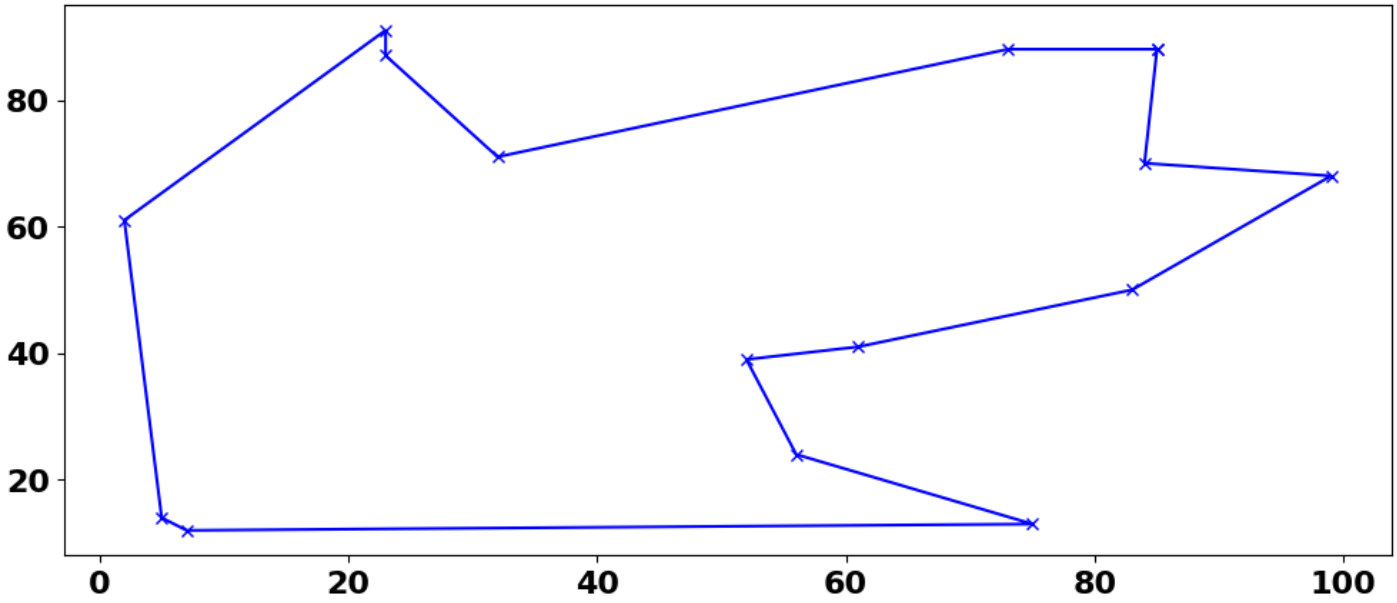
In [25]:
```
from itertools import product
from mip import Model, xsum, minimize, BINARY
# adapted from https://python-mip.readthedocs.io/en/latest/examples.html

# number of nodes and list of vertices
n, V = len(cities), set(range(len(cities)))

# distances matrix
c = distance_matrix

model = Model()
```

```
# binary variables indicating if arc (i,j) is used on the route or not
x = [[model.add_var(var_type=BINARY) for j in V] for i in V]

# continuous variable to prevent subtours: each city will have a
# different sequential id in the planned route except the first one
y = [model.add_var() for i in V]

# objective function: minimize the distance
model.objective = minimize(xsum(c[i][j]*x[i][j] for i in V for j in V))

# constraint : leave each city only once
for i in V:
    model += xsum(x[i][j] for j in V - {i}) == 1

# constraint : enter each city only once
for i in V:
    model += xsum(x[j][i] for j in V - {i}) == 1

# subtour elimination
for (i, j) in product(V - {0}, V - {0}):
    if i != j:
        model += y[i] - (n+1)*x[i][j] >= y[j]-n

# optimizing
model.optimize(max_seconds=30)

# checking if a solution was found
if model.num_solutions:
    tour = []
    print(
        'route with total distance %g found'
        % model.objective_value
    )
    nc = 0
    while True:
        nc = [i for i in V if x[nc][i].x >= 0.99][0]
        tour.append(nc)
        #print(' -> %s' % places[nc])
        if nc == 0:
            break
```

```
route with total distance 361.005 found
```

In [26]:

```
plot_solution(cities, tour)
```



In [ ]: